2.25

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Exercise 2.25 Data Loading

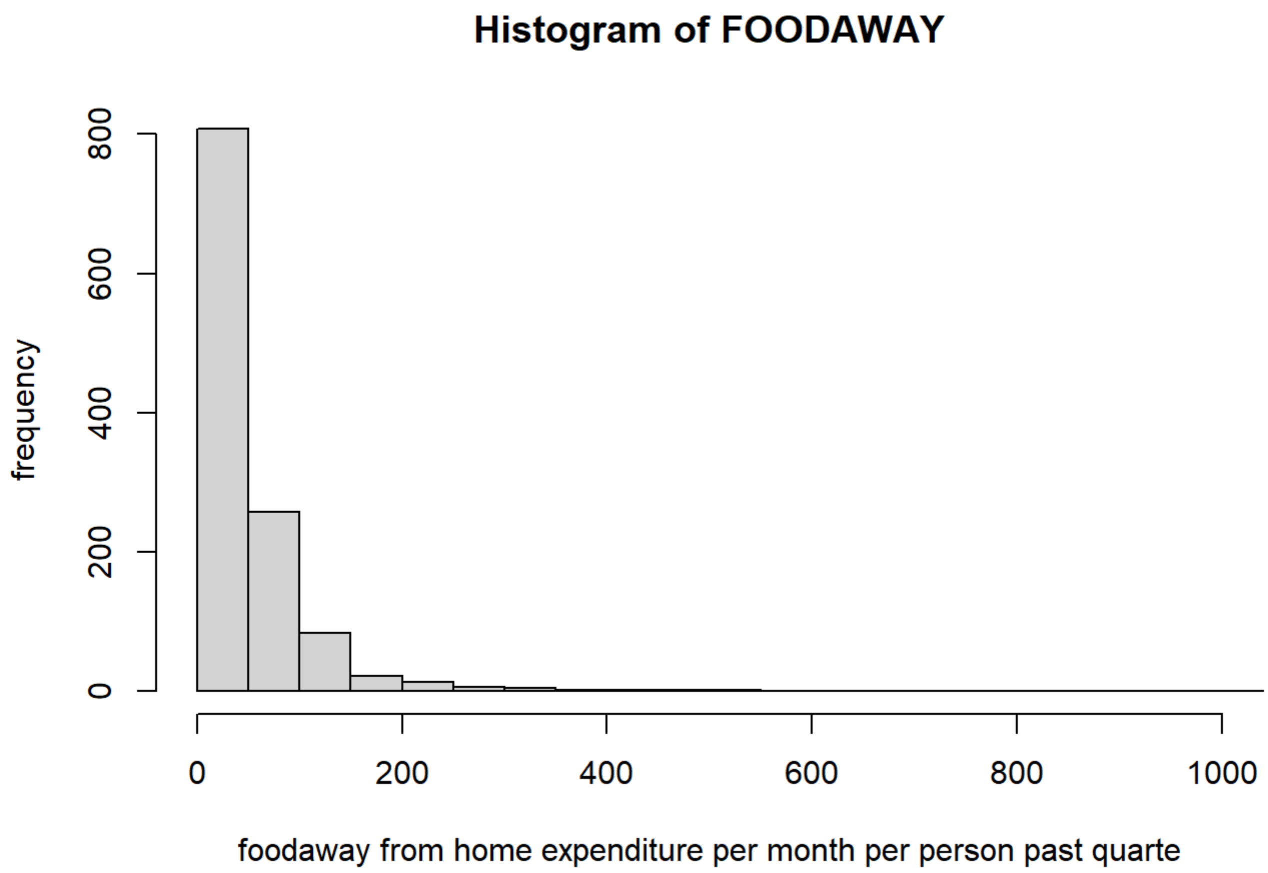
```
url <- "http://www.principlesofeconometrics.com/poe5/data/rdata/cex5_small.rdata"
dest <- file.path(tempdir(), "cex5_small.rdata")
download.file(url, dest, mode = "wb")
load(dest)
```

Part (a)

```
##a.
cat(
  "Mean of FOODWAY:", mean(cex5_small$foodaway), "\n",
  "Median of FOODWAY:", median(cex5_small$foodaway), "\n",
  "25th percentiles of FOODWAY:", quantile(cex5_small$foodaway, 0.25), "\n",
  "75th percentiles of FOODWAY:", quantile(cex5_small$foodaway, 0.75), "\n",
  sep = " "
)
```

```
## Mean of FOODWAY: 49.27085
##  Median of FOODWAY: 32.555
##  25th percentiles of FOODWAY: 12.04
##  75th percentiles of FOODWAY: 67.5025
```

```
hist(cex5_small$foodaway,xlab = 'foodaway from home expenditure per month per person past qua
rte', ylab = 'frequency', main = 'Histogram of FOODAWAY',
     xlim = c(0,1000), breaks= 20)
```



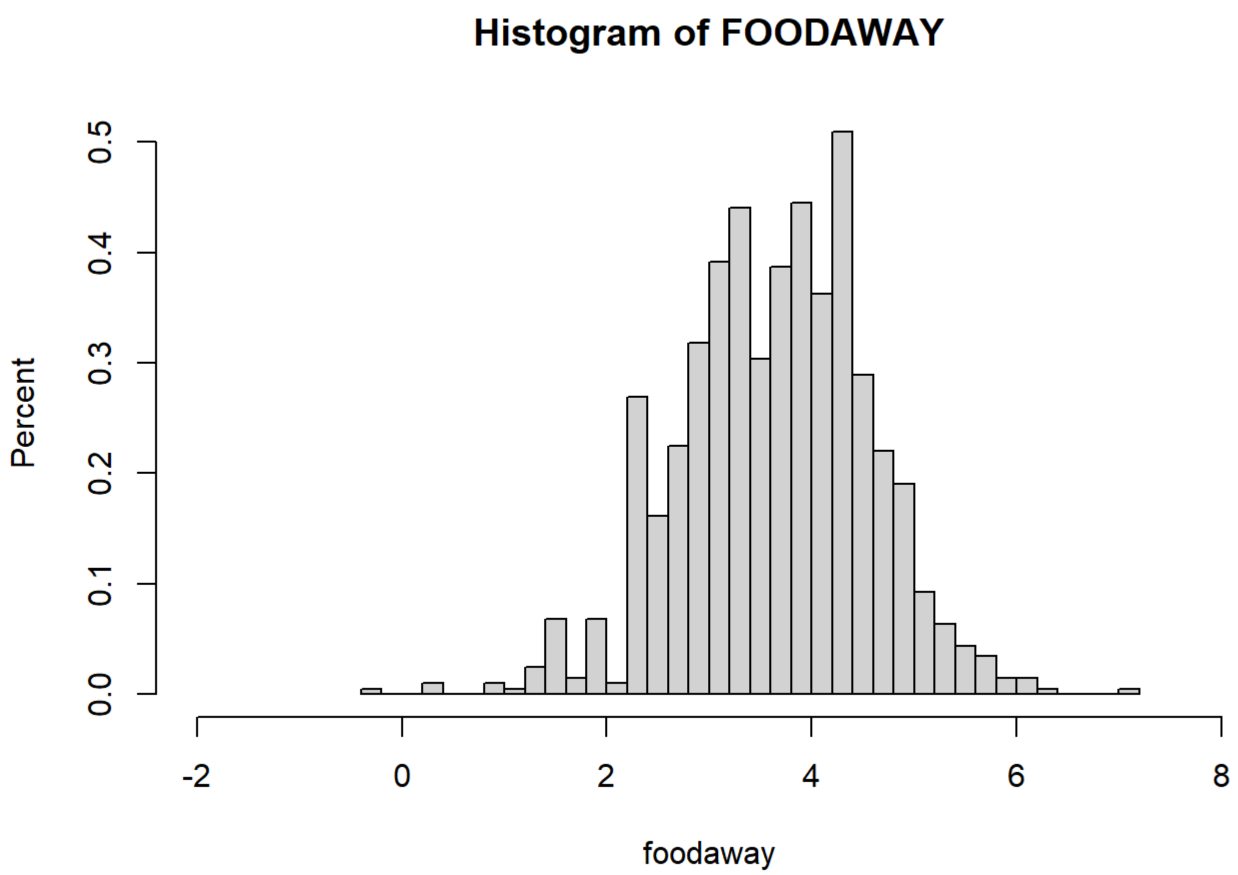
Part (b)

```
##b.
cat(
  "advanced = 1:", "\n",
  "N:", sum(cex5_small$advanced), "\n",
  "Mean:", mean(cex5_small$foodaway[cex5_small$advanced == 1]), "\n",
  "Median:", median(cex5_small$foodaway[cex5_small$advanced == 1]), "\n",
  "college = 1:", "\n",
  "N:", sum(cex5_small$college), "\n",
  "Mean:", mean(cex5_small$foodaway[cex5_small$college == 1]), "\n",
  "Median:", median(cex5_small$foodaway[cex5_small$college == 1]), "\n",
  "None:", "\n",
  "N:", sum(cex5_small$advanced == 0 & cex5_small$college == 0), "\n",
  "Mean:", mean(cex5_small$foodaway[cex5_small$advanced == 0 & cex5_small$college == 0]),
"\n",
  "Median:", median(cex5_small$foodaway[cex5_small$advanced == 0 & cex5_small$college == 0]),
"\n",
  sep = ""
)
```

```
## advanced = 1:
## N:257
## Mean:73.15494
## Median:48.15
## college = 1:
## N:369
## Mean:48.59718
## Median:36.11
## None:
## N:574
## Mean:39.01017
## Median:26.02
```

Part (c)

```
##c.
hist(log(cex5_small$foodaway),prob = TRUE,xlab = 'foodaway', ylab = 'Percent', main = 'Histog
ram of FOODAWAY',
     xlim = c(-2,8), breaks= 30)
```




```
cat(
  "Number of values loss after take log is:",sum(!is.finite(log(cex5_small$foodaway))), "\n",
  "Since those sample are 0, ln(0) is undefined and become missing value",
  sep = ""
)
```

```
## Number of values loss after take log is:178
## Since those sample are 0, ln(0) is undefined and become missing value
```

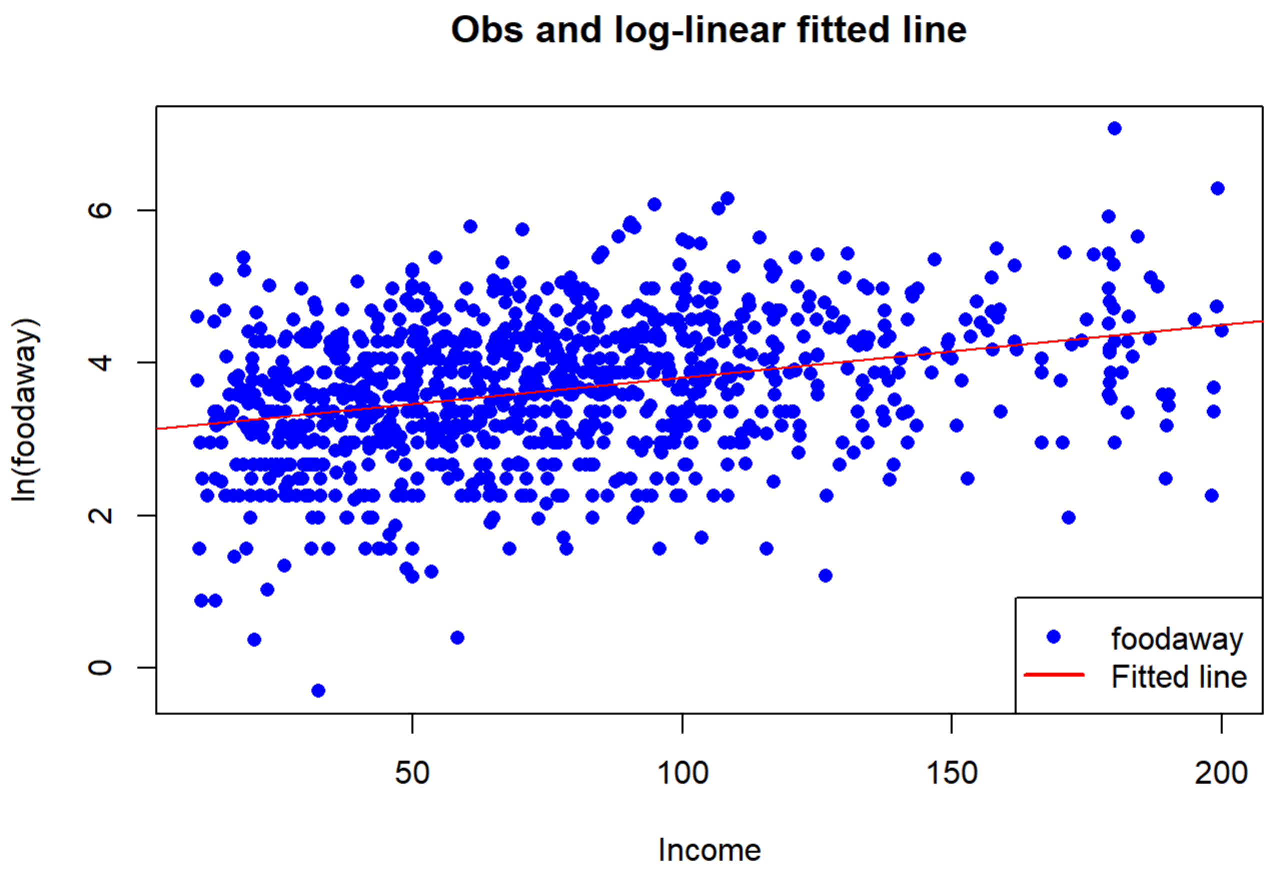
Part (d)

```
##d.
idx <- !is.na(cex5_small$foodaway) & cex5_small$foodaway > 0 &
  !is.na(cex5_small$income)

df <- cex5_small[idx, ]

mod1 <- lm(log(foodaway) ~ income, data = df)
intercept <- coef(mod1)[1]
slope     <- coef(mod1)[2]
cat(
  "intercept is: ",intercept, "\n",
  "slope is: ", slope, "\n",
  "We estimate that each additional $100 household income increases food away expenditures",
"\n",
  "per person of about 0.69%, other factors held constant", "\n",
  sep=""
)
```

```
## intercept is: 3.1293
## slope is: 0.006901748
## We estimate that each additional $100 household income increases food away expenditures
## per person of about 0.69%, other factors held constant
```

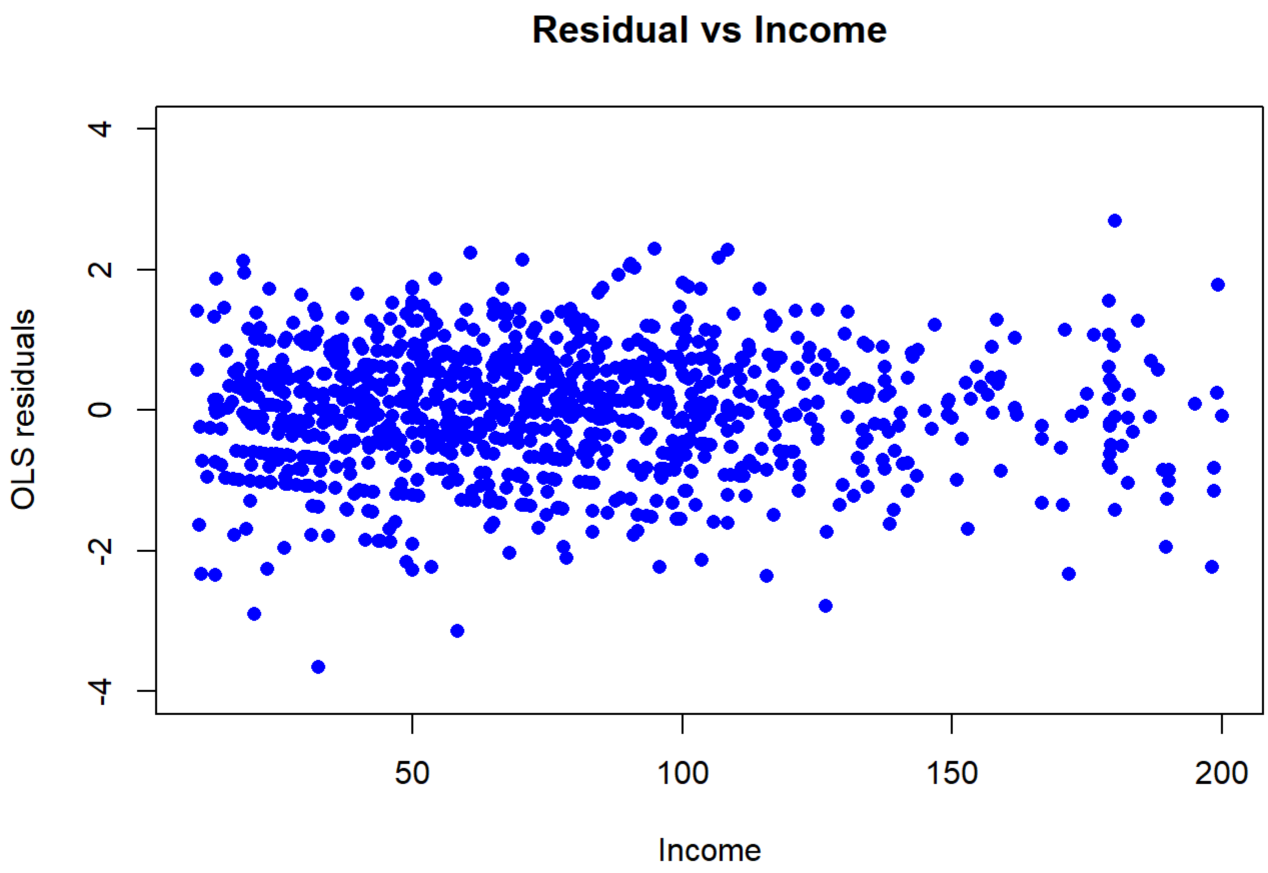
Part (e)

```
##e.
lnfood = log(cex5_small$foodaway)
plot(cex5_small$income, lnfood,
     xlab="Income",
     ylab="ln(foodaway)",
     main = "Obs and log-linear fitted line",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept,slope ,col = 'red')
legend("bottomright",legend = c("foodaway", "Fitted line"),
       col    = c("blue", "red"),pch    = c(16, NA),     lty    = c(NA, 1),
       lwd    = c(NA, 2))
```



Part (f)

```
##f.
res = resid(mod1)
plot(df$income,res,
     xlab = "Income",ylab = "OLS residuals",
     main = "Residual vs Income",
     ylim = c(-4,4),
     pch =16,col = 'blue',
     type='p')
```


Residual vs Income
OLS residuals
Income
4
2
0
-2
-4
50
100
150
200